AP BIOLOGY · UNIT 1

TOPIC 1.6

Nucleic Acids: DNA & RNA

Structure, function, and the molecular language of life

Learning Objective: Describe the structure and function of DNA and RNA — the molecules that store, transmit, and express biological information in all living organisms.

CHAPTER 1

Why Nucleic Acids Matter

Every cell in your body contains nucleic acids — the information-carrying molecules that direct everything from protein synthesis to heredity. Without them, life as we know it simply couldn't exist. Understanding nucleic acids is foundational to virtually every other topic in AP Biology, from gene expression to evolution.

The Big Idea

In nucleic acids — both **DNA** (deoxyribonucleic acid) and **RNA** (ribonucleic acid) — **biological information is encoded in sequences of nucleotide monomers**. Think of nucleotides as letters in an alphabet. Just as the English alphabet uses 26 letters to write millions of words, nucleic acids use a small set of nucleotide "letters" to encode the instructions for building and running an organism.

DNA serves as the **long-term information storage** molecule — the master blueprint of the cell. RNA acts as a **versatile messenger and worker**, carrying instructions from DNA to the ribosome and playing key roles in protein synthesis. Together, they form the central dogma of molecular biology: **DNA → RNA → Protein**.

🎯 **AP Exam Note:** You do NOT need to memorize the detailed molecular structure of individual nucleotides, but you DO need to know the three structural components and how nucleotides connect to form polynucleotide chains.

Nucleotide Structure: The Building Blocks

Nucleotides are the **monomers** (individual subunits) of nucleic acids. Every nucleotide — whether it's part of DNA or RNA — is built from the same three structural components. Understanding these components is critical for the AP Exam.

Five-Carbon Sugar

Either **deoxyribose** (in DNA) or **ribose** (in RNA). The key difference is a single oxygen atom — ribose has an –OH group on its 2' carbon, while deoxyribose has only an –H. This "missing" oxygen is where the "deoxy" prefix comes from. The sugar forms the backbone of the nucleic acid strand and determines which type of nucleic acid it belongs to.

Phosphate Group

A phosphorus atom bonded to four oxygen atoms (PO_4^{3-}). The phosphate group is attached to the **5' carbon** of the sugar. Phosphate groups are what give nucleic acids their **negative charge** and form the covalent **phosphodiester bonds** that link one nucleotide to the next in the sugar-phosphate backbone.

Nitrogenous Base

The information-carrying part of the nucleotide, attached to the **1' carbon** of the sugar. There are five possible bases: **Adenine (A), Thymine (T), Guanine (G), Cytosine (C),** and **Uracil (U)**. DNA uses A, T, G, C. RNA uses A, U, G, C. The **sequence** of these bases encodes genetic information.

The Five Nitrogenous Bases

Nitrogenous bases are classified into two categories based on their ring structure. Knowing which bases belong to which group — and which nucleic acid uses which bases — is essential for understanding base pairing and the structural differences between DNA and RNA.

Purines (Double Ring)

Adenine (A) — Found in both DNA and RNA. Pairs with thymine in DNA and uracil in RNA.

Guanine (G) — Found in both DNA and RNA. Always pairs with cytosine.

💡 **Memory tip:** Purines are "pure as gold" — **A** and **G**. They have a larger, double-ring structure.

Pyrimidines (Single Ring)

Cytosine (C) — Found in both DNA and RNA. Always pairs with guanine.

Thymine (T) — Found **only in DNA**. Pairs with adenine.

Uracil (U) — Found **only in RNA**. Replaces thymine and pairs with adenine.

💡 **Memory tip:** Pyrimidines — **C**, **T**, **U** — "CUT the pie" (py-rimidine).

🧬 **Key Rule:** A purine always pairs with a pyrimidine. This keeps the width of the double helix consistent — one big ring + one small ring = uniform diameter. Purine + purine would be too wide; pyrimidine + pyrimidine would be too narrow.

Directionality: The 5' and 3' Ends

Nucleic acids are not just random chains — they have a defined **directionality** that is critical for understanding how DNA is replicated and how RNA is transcribed. Every strand of DNA or RNA has two chemically distinct ends:

5' End (Five Prime)

The end of the strand where a free **phosphate group** is attached to the 5' carbon of the terminal sugar. This is the "beginning" of the strand. New nucleotides are NOT added here during synthesis.

Phosphodiester Bonds

Covalent bonds linking the 3' carbon of one nucleotide's sugar to the 5' phosphate of the next. These bonds form the strong **sugar-phosphate backbone** that gives the strand its structural integrity.

3' End (Three Prime)

The end of the strand where a free **hydroxyl group (–OH)** is attached to the 3' carbon of the terminal sugar. This is where **new nucleotides are added** during DNA replication and RNA transcription.

> **During nucleic acid synthesis, nucleotides are always added to the 3' end of the growing strand.** This means the strand grows in the **5' → 3' direction**. This is a universal rule for all DNA and RNA polymerases and is one of the most frequently tested concepts on the AP Exam.

How Nucleotides Join Together

When nucleotides polymerize to form a nucleic acid strand, a **dehydration synthesis reaction** (also called a condensation reaction) creates a **phosphodiester bond** between adjacent nucleotides. This process is catalyzed by enzymes — DNA polymerase for DNA and RNA polymerase for RNA.

The resulting polymer is called a **polynucleotide**. The sugar-phosphate backbone is the structural scaffold — it's the same in every strand regardless of sequence. The **nitrogenous bases project inward** (in DNA) or outward (in single-stranded RNA), and their specific sequence is what carries the genetic information. Think of the backbone as the binding of a book and the bases as the words on the pages.

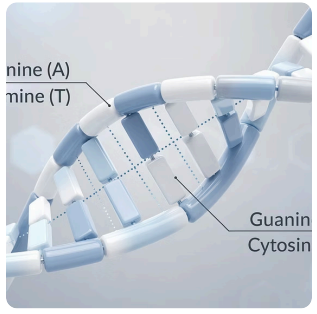
⚡ **Energy Connection:** Each incoming nucleotide arrives as a nucleoside triphosphate (like ATP). The hydrolysis of the two extra phosphates provides the energy needed to drive the bond formation. This is why nucleic acid synthesis requires energy!

The DNA Double Helix

DNA's structure — famously elucidated by Watson and Crick in 1953 using data from Rosalind Franklin — is one of the most iconic shapes in all of biology. Understanding its architecture is essential for grasping how DNA stores information, replicates itself, and is read by the cell.

Antiparallel Double Helix

DNA consists of **two polynucleotide strands** wound around each other in a right-handed helix. The two strands run in **opposite directions** — one strand runs 5' → 3' while the complementary strand runs 3' → 5'. This antiparallel arrangement is crucial for DNA replication and is a key concept for the AP Exam. The sugar-phosphate backbones form the outer "rails" of the helix, while the nitrogenous bases face inward like "rungs of a ladder."



Complementary Base Pairing

The two strands are held together by **hydrogen bonds** between complementary base pairs. **Adenine (A) pairs with Thymine (T)** via **2 hydrogen bonds**, and **Cytosine (C) pairs with Guanine (G)** via **3 hydrogen bonds**. Because G-C pairs have more hydrogen bonds, regions rich in G-C content are more stable and harder to separate (denature). This complementary pairing means that if you know the sequence of one strand, you can determine the other.

RNA: The Versatile Single Strand

While DNA serves as the stable, long-term repository of genetic information, RNA is the **dynamic, versatile** molecule that carries out many of the cell's day-to-day functions. RNA comes in several forms — messenger RNA (mRNA), transfer RNA (tRNA), ribosomal RNA (rRNA), and many others — each with specialized roles in gene expression.

Key Structural Features of RNA

1

Single-Stranded

Unlike DNA's double helix, RNA is **typically single-stranded**. However, RNA can fold back on itself to form complex three-dimensional structures through intramolecular base pairing (like hairpin loops and stem-loops). These shapes are critical for RNA function — for example, tRNA's cloverleaf shape allows it to carry amino acids to the ribosome.

2

Ribose Sugar

RNA contains the sugar **ribose**, which has a hydroxyl group (–OH) on the 2' carbon. This extra –OH makes RNA more chemically reactive and less stable than DNA — which is one reason cells use DNA (not RNA) for long-term information storage.

3

Uracil Replaces Thymine

In RNA, the base **uracil (U)** replaces thymine. Uracil pairs with adenine (**A-U**) via hydrogen bonds, just as thymine pairs with adenine in DNA. The base pairing rule for RNA: **A-U** and **C-G**.

DNA vs. RNA: Side-by-Side Comparison

Understanding the structural and functional differences between DNA and RNA is one of the most frequently tested concepts in AP Biology Unit 1. Use the table below as a quick-reference study tool, then review the visual summary underneath.

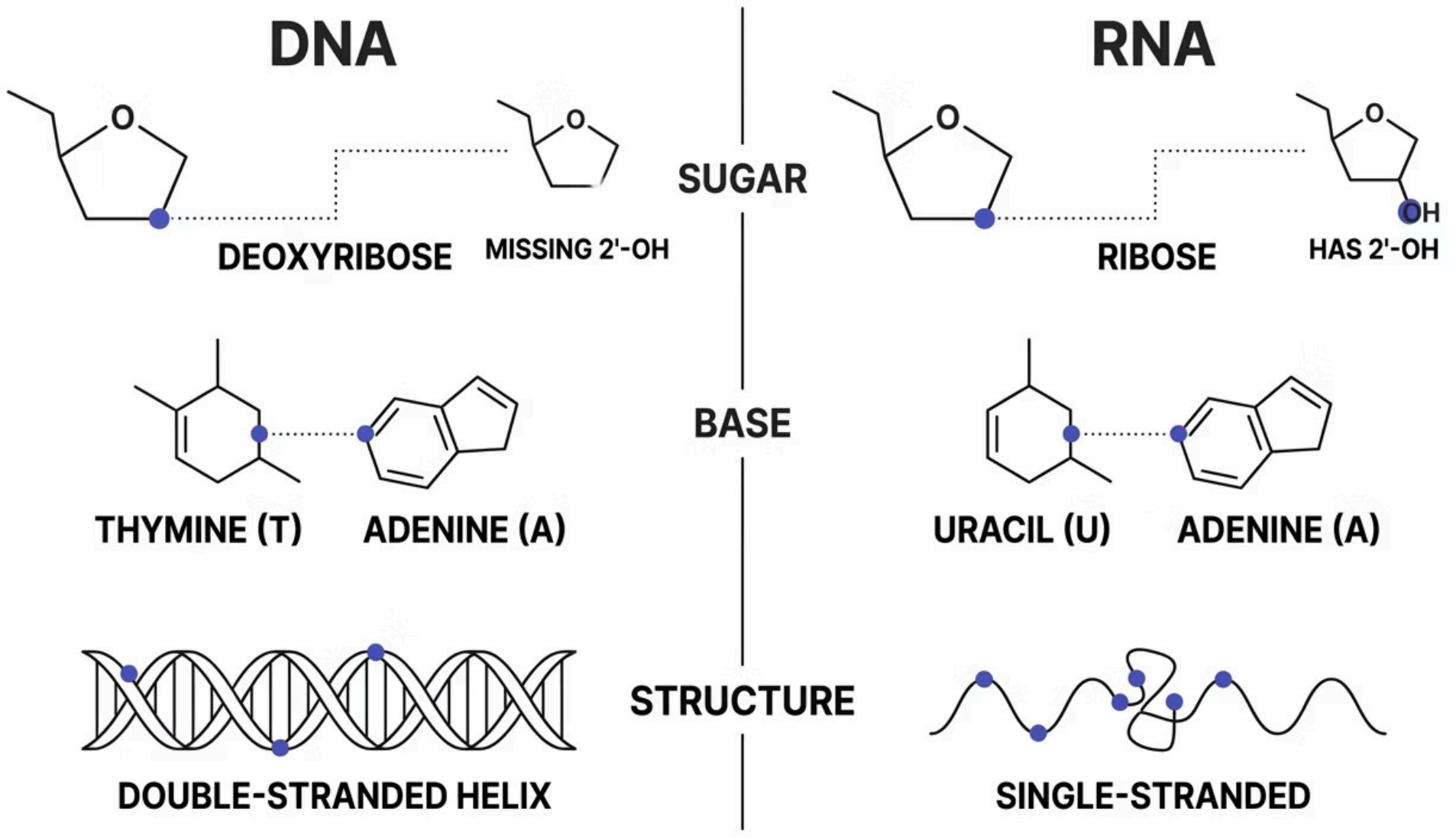
Feature	DNA	RNA
Full Name	Deoxyribonucleic acid	Ribonucleic acid
Sugar	Deoxyribose (–H on 2' carbon)	Ribose (–OH on 2' carbon)
Nitrogenous Bases	A, T, G, C	A, U, G, C
Unique Base	Thymine (T)	Uracil (U)
Structure	Double-stranded (antiparallel helix)	Typically single-stranded
Base Pairing	A-T (2 H-bonds), C-G (3 H-bonds)	A-U, C-G (intramolecular)
Stability	More stable (no 2' –OH, double-stranded)	Less stable (2' –OH, single-stranded)
Primary Function	Long-term genetic information storage	Gene expression (mRNA, tRNA, rRNA)
Location (Eukaryotes)	Primarily nucleus	Nucleus and cytoplasm

🔑 Three Key Differences

1. Sugar: Deoxyribose vs. Ribose
2. Base: Thymine vs. Uracil
3. Strands: Double vs. Single

Visualizing the Three Key Differences

Let's break down each structural difference between DNA and RNA with clear visual comparisons. These three distinctions are testable on the AP Exam and appear in both multiple-choice and free-response questions.



Why Does It Matter?

These structural differences aren't random — they reflect the **functional roles** of each molecule. DNA's double-stranded structure and deoxyribose sugar make it exceptionally **stable**, perfect for long-term storage of genetic information across generations. RNA's single-stranded structure and ribose sugar make it more **flexible and reactive** — ideal for its temporary roles in gene expression, where it needs to be made quickly, used, and then degraded.

Common AP Exam Traps

- Don't confuse thymine with uracil — T is DNA only, U is RNA only
- Remember that **both** DNA and RNA contain adenine, guanine, and cytosine
- RNA **can** form double-stranded regions through self-folding, but is **typically** single-stranded
- The antiparallel nature of DNA means strands run in **opposite** directions

Base Pairing Rules: The Heart of the Code

Complementary base pairing is the molecular foundation of DNA replication, transcription, and translation. Mastering these rules is non-negotiable for AP Biology success.

DNA Base Pairing

A — T (2 hydrogen bonds)

C — G (3 hydrogen bonds)

These pairs hold the two antiparallel strands of the double helix together. The consistent purine-pyrimidine pairing maintains a uniform helix width of approximately 2 nm.

RNA Base Pairing

A — U (2 hydrogen bonds)

C — G (3 hydrogen bonds)

When RNA is transcribed from DNA, adenine in the DNA template pairs with uracil in the new RNA strand. RNA can also base-pair with itself when it folds into secondary structures.

Practice: Complementary Strand Writing

If a DNA strand reads: **5'— A T G C C T A A G —3'**

The complementary DNA strand is: **3'— T A C G G A T T C —5'** (remember antiparallel!)

The mRNA transcribed from the template strand would be: **5'— A U G C C U A A G —3'** (U replaces T)

Key Takeaways for the AP Exam

Here's everything you need to remember about Topic 1.6 Nucleic Acids, distilled into the essential knowledge points that appear most frequently on the AP Biology exam.

01

Nucleotide = Sugar + Phosphate + Base

Every nucleotide monomer has three parts: a five-carbon sugar (deoxyribose or ribose), a phosphate group, and one of five nitrogenous bases (A, T, G, C, or U). The sequence of bases encodes biological information.

02

Strands Have Directionality (5' → 3')

Nucleic acid strands have defined 5' (phosphate) and 3' (hydroxyl) ends. New nucleotides are always added to the 3' end during synthesis. Strands grow 5' to 3'.

03

DNA = Antiparallel Double Helix

Two strands run in opposite directions, held together by hydrogen bonds between complementary bases: A-T (2 H-bonds) and C-G (3 H-bonds).

04

RNA Differs in Three Ways

RNA uses ribose (not deoxyribose), uracil (not thymine), and is typically single-stranded (not double-stranded). In RNA, A pairs with U.

05

Information Is in the Sequence

The specific linear sequence of nucleotide bases is what encodes genetic information — not the backbone, not the sugar, not the phosphate. The backbone is structurally identical regardless of sequence.

📝 **Scope Reminder:** The molecular structure of specific nucleotides (ring structures, bond angles, etc.) is **beyond the scope** of the AP Exam. Focus on the three components, base pairing rules, directionality, and DNA vs. RNA differences.